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SICHUAN UNIVERSITY

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Mathematical Programming

An Introduction to Optimization

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Course Scope and References

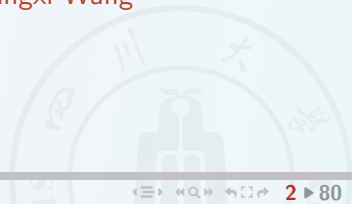
Course scope

- Mathematical programming models and terminology
- Linear-algebra foundations for optimization
- Affine and convex geometry
- Linear programming and polyhedral structure

Primary references

- Melvyn W. Jeter, *Mathematical Programming: An Introduction to Optimization*
- Stephen Boyd and Lieven Vandenberghe, *Convex Optimization*

Instructor: Changxi Wang



Learning Objectives

By the end of this unit, students should be able to:

- translate a verbal decision problem into variables, objectives, and constraints;
- distinguish linear, nonlinear, continuous, and discrete models;
- use linear-algebra and topology concepts in optimization arguments;
- characterize affine sets, convex sets, polyhedra, and polytopes;
- explain why linear-program feasible regions are convex.



Optimization Turns Decisions into Mathematical Models

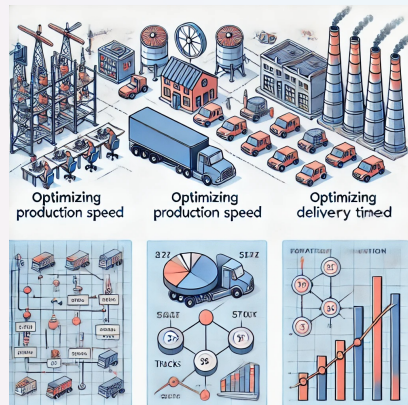
- This course assumes prior study of linear algebra, calculus, and multivariable calculus.
- Optimization selects the best feasible decision according to a stated objective.
- Representative applications include:
 - economic analysis and management;
 - production planning and logistics;
 - engineering design and resource allocation;
 - scheduling, networks, and operations research.



Optimization Coordinates Production Decisions

- production quantity;
- production speed;
- inventory and delivery timing;
- allocation of limited resources.

A model makes the trade-offs explicit and testable.



Optimization Supports Transportation and Logistics

- vehicle routing;
- public-transport planning;
- delivery-time control;
- network design;
- capacity allocation.



A Mathematical Program Has an Objective and a Feasible Set

Generic formulation

$$\min_{x \in \Omega} f(x) \quad \text{or} \quad \max_{x \in \Omega} f(x), \quad x = (x_1, \dots, x_n)^\top.$$

- $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the objective (or cost) function.
- $\Omega \subseteq \mathbb{R}^n$ is the feasible set.
- The problem is **unconstrained** when $\Omega = \mathbb{R}^n$.
- It is **constrained** when feasibility restricts the decision vector.



A Linear Program Uses Linear Objectives and Constraints

- Linear objective:

$$f(x) = c_1x_1 + \cdots + c_nx_n = c^T x.$$

- Linear equality: $a_1x_1 + \cdots + a_nx_n = b$.
- Linear inequality: $a_1x_1 + \cdots + a_nx_n \leq b$ or $\geq b$.
- If the objective or any constraint is nonlinear, the model is a nonlinear program.

Key distinction

Linearity concerns the mathematical expressions, not the application domain.

Example 1: Production Planning Starts from Resource Data

A manufacturer produces products $P_1, \dots, P_n$ using raw materials $M_1, \dots, M_m$.

	P_1	P_2	$\dots$	P_n
M_1	a_{11}	a_{12}	$\dots$	a_{1n}
M_2	a_{21}	a_{22}	$\dots$	a_{2n}
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
M_m	a_{m1}	a_{m2}	$\dots$	a_{mn}

- b_i : available units of material M_i ;
- c_j : revenue per unit of product P_j ;
- goal: maximize total daily revenue.



Example 1: Variables Connect Production to Revenue and Usage

- Decision variable: x_j = number of units of product P_j produced.
- Total revenue:

$$c_1x_1 + \cdots + c_nx_n = \sum_{j=1}^n c_jx_j.$$

- Material M_i consumed:

$$a_{i1}x_1 + \cdots + a_{in}x_n = \sum_{j=1}^n a_{ij}x_j.$$

- Capacity requirement: $\sum_{j=1}^n a_{ij}x_j \leq b_i$ for every i .



Example 1: The Production Model Is a Linear Program

$$\begin{aligned} \max_x \quad & \sum_{j=1}^n c_j x_j \\ \text{subject to} \quad & \sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, \dots, m, \\ & x_j \geq 0, \quad j = 1, \dots, n. \end{aligned}$$

In matrix form:

$$\max_{x \geq 0} c^T x \quad \text{subject to} \quad Ax \leq b.$$



Variable Domains and Slack Variables Change the Model Form

- A **continuous variable** may take any real value in its allowed interval.
- A **discrete variable** takes values from a discrete set, often $\mathbb{Z}$ or $\{0, 1\}$.
- A **slack variable** converts a $\leq$ inequality to an equality:

$$a_{11}x_1 + \cdots + a_{1n}x_n \leq b_1 \iff a_{11}x_1 + \cdots + a_{1n}x_n + s_1 = b_1, \quad s_1 \geq 0.$$

- A slack variable has objective coefficient zero unless the model assigns it a separate cost.



Slack Variables Produce Equality Standard Form

$$\begin{aligned} \max \quad & \sum_{j=1}^n c_j x_j + \sum_{i=1}^m 0 s_i \\ \text{subject to} \quad & \sum_{j=1}^n a_{ij} x_j + s_i = b_i, \quad i = 1, \dots, m, \\ & x_j \geq 0, \quad j = 1, \dots, n, \\ & s_i \geq 0, \quad i = 1, \dots, m. \end{aligned}$$

With $s = b - Ax$, the equality form is feasible exactly when the original inequalities are feasible.

Example 2: Assignment Uses a Productivity Matrix

Four salespeople must be assigned to four sales districts.

	D_1	D_2	D_3	D_4
S_1	5	10	8	2
S_2	3	4	2	6
S_3	9	5	4	5
S_4	5	5	6	4

The goal is to choose a one-to-one assignment that maximizes total productivity.

Example 2: Binary Variables Encode an Assignment

Define

$$x_{ij} = \begin{cases} 1, & \text{if salesperson } S_i \text{ is assigned to district } D_j, \\ 0, & \text{otherwise.} \end{cases}$$

If r_{ij} is the productivity rating, the total value is

$$\sum_{i=1}^4 \sum_{j=1}^4 r_{ij} x_{ij}.$$

One-to-one assignment requires

$$\sum_{j=1}^4 x_{ij} = 1 \quad \forall i, \quad \sum_{i=1}^4 x_{ij} = 1 \quad \forall j.$$

Example 2: Compact Assignment Formulation

$$\begin{aligned} \max_x \quad & \sum_{i=1}^4 \sum_{j=1}^4 r_{ij} x_{ij} \\ \text{subject to} \quad & \sum_{j=1}^4 x_{ij} = 1, \quad i = 1, \dots, 4, \\ & \sum_{i=1}^4 x_{ij} = 1, \quad j = 1, \dots, 4, \\ & x_{ij} \in \{0, 1\}, \quad i, j = 1, \dots, 4. \end{aligned}$$

Integrality

For the assignment constraint matrix, the LP relaxation also has integral extreme points; the binary model states the intended decisions explicitly.

Example 3: Production-Line Assignment

Three production lines can make flour brands A and B. Entries are ten-pound sacks per day.

	L_1	L_2	L_3
A	15000	10000	12000
B	12000	13500	14500

Decision question

Assign each line to exactly one brand to maximize total daily output.

Example 3: Two Binary Indicators per Production Line

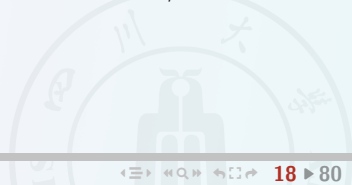
For each line L_i , define

$$x_i = \begin{cases} 1, & L_i \text{ makes A,} \\ 0, & \text{otherwise,} \end{cases} \quad w_i = \begin{cases} 1, & L_i \text{ makes B,} \\ 0, & \text{otherwise.} \end{cases}$$

Complementarity can be written as

$$x_i w_i = 0, \quad i = 1, 2, 3.$$

Together with $x_i, w_i \geq 0$, this prevents simultaneous assignment to both brands, but it does not yet force a line to be used.



Example 3: Exact Assignment Makes Complementarity Redundant

Requiring every line to make exactly one brand gives

$$x_i + w_i = 1, \quad i = 1, 2, 3,$$

with $x_i, w_i \in \{0, 1\}$.

- These constraints already imply $x_i w_i = 0$.
- The single aggregate condition $\sum_{i=1}^3 x_i w_i = 0$ is equivalent only under nonnegativity.
- The binary linear formulation is simpler than retaining a nonlinear complementarity constraint.



Example 3: A Complementarity Formulation Is Nonlinear

$$\begin{aligned} \max_{x,w} \quad & 15000x_1 + 10000x_2 + 12000x_3 \\ & + 12000w_1 + 13500w_2 + 14500w_3 \\ \text{subject to} \quad & x_i + w_i = 1, & i = 1, 2, 3, \\ & \sum_{i=1}^3 x_i w_i = 0, \\ & 0 \leq x_i, w_i \leq 1, & i = 1, 2, 3. \end{aligned}$$

The product terms make this formulation nonlinear. With binary variables, the product constraint can be removed.

Example 4: Quadratic Programming Adds Pairwise Terms

A quadratic program may take the form

$$\begin{aligned} \max_x \quad & c^\top x + x^\top Bx \\ \text{subject to} \quad & Ax = b, \\ & x \geq 0. \end{aligned}$$

Equivalently,

$$x^\top Bx = \sum_{i=1}^n \sum_{j=1}^n b_{ij} x_i x_j.$$

Convexity or concavity depends on the quadratic matrix and on whether the problem minimizes or maximizes.

Example 4: Portfolio Selection Trades Return for Risk

$$\begin{aligned} \max_x \quad & c^\top x - x^\top Bx \\ \text{subject to} \quad & 1^\top x = 1, \\ & x \geq 0. \end{aligned}$$

- x_i : fraction invested in asset i ;
- c_i : expected return of asset i ;
- $x^\top Bx$: quadratic risk measure;
- the minus sign penalizes risk while rewarding expected return.



Nearby Neighborhoods Formalize What “Local” Means

For $x, \bar{x} \in \mathbb{R}^n$, the Euclidean distance is

$$\|x - \bar{x}\|_2 = \sqrt{\sum_{i=1}^n (x_i - \bar{x}_i)^2}.$$

The open ε -neighborhood of $\bar{x}$ is

$$N_\varepsilon(\bar{x}) = \{x \in \mathbb{R}^n : \|x - \bar{x}\|_2 < \varepsilon\}, \quad \varepsilon > 0.$$

Local optimality compares feasible points only inside such a neighborhood.



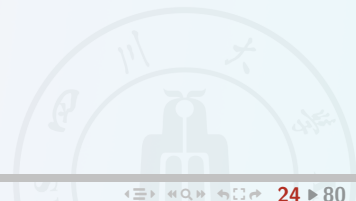
Local and Global Minima Differ by Comparison Set

Let $\bar{x} \in \Omega$.

- **Local minimum:** there exists $\varepsilon > 0$ such that

$$f(\bar{x}) \leq f(x) \quad \forall x \in \Omega \cap N_\varepsilon(\bar{x}).$$

- **Strict local minimum:** replace $\leq$ by $<$ for every feasible $x \neq \bar{x}$ in that neighborhood.
- **Global minimum:** $f(\bar{x}) \leq f(x)$ for all $x \in \Omega$.
- **Strict global minimum:** $f(\bar{x}) < f(x)$ for all $x \in \Omega \setminus \{\bar{x}\}$.



Interior, Open Sets, and Exterior

$$\begin{aligned}\text{Int}(\Omega) &= \{x \in \Omega : \exists \varepsilon > 0, N_\varepsilon(x) \subseteq \Omega\}, \\ \text{Ext}(\Omega) &= \{x \in \mathbb{R}^n : \exists \varepsilon > 0, N_\varepsilon(x) \subseteq \mathbb{R}^n \setminus \Omega\}.\end{aligned}$$

- Ω is open when $\Omega = \text{Int}(\Omega)$.
- Every point of an open set has a sufficiently small neighborhood contained in the set.
- The exterior is the interior of the complement.



Boundary and Accumulation Points Describe Set Limits

- x is a **boundary point** of Ω when every neighborhood intersects both Ω and $\mathbb{R}^n \setminus \Omega$.
- The boundary is

$$\partial\Omega = \overline{\Omega} \setminus \text{Int}(\Omega).$$

- Ω is closed when it contains all of its boundary points, equivalently $\overline{\Omega} = \Omega$.
- x is an **accumulation point** if

$$\forall \varepsilon > 0, \quad (N_\varepsilon(x) \setminus \{x\}) \cap \Omega \neq \emptyset.$$



Sensitivity Analysis Reuses Structure after Data Change

Model data are estimates or measurements and may change over time.

Existing problem

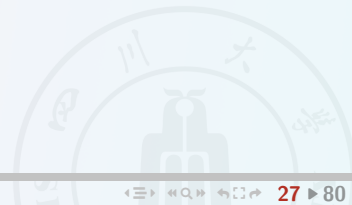
Known data, feasible set, and optimal solution



Modified problem

Updated data and a revised optimal solution

Post-optimal analysis asks whether the original solution or factorization can be updated without solving from scratch.



Example 5: Transportation Balances Supply and Demand

- Products are stored at m origins and sold at n outlets.
- c_{ij} is the unit shipping cost from origin i to outlet j .
- Origin i has s_i units available.
- Outlet j requires exactly d_j units.
- Decision variable x_{ij} is the quantity shipped from i to j .

Goal

Meet all outlet demands at minimum total shipping cost without exceeding any origin's supply.



Example 5: Transportation Model

$$\begin{aligned} \min_x \quad & \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \\ \text{subject to} \quad & \sum_{j=1}^n x_{ij} \leq s_i, \quad i = 1, \dots, m, \\ & \sum_{i=1}^m x_{ij} = d_j, \quad j = 1, \dots, n, \\ & x_{ij} \geq 0, \quad i = 1, \dots, m, \quad j = 1, \dots, n. \end{aligned}$$

Feasibility requires total available supply to be at least total demand.

Parametric Programming Models Demand Growth

Suppose outlet j has current demand d_j and daily growth $\bar{d}_j$. After $t \geq 0$ days:

$$\begin{aligned} \min_x \quad & \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \\ \text{subject to} \quad & \sum_{j=1}^n x_{ij} \leq s_i, \quad i = 1, \dots, m, \\ & \sum_{i=1}^m x_{ij} = d_j + t\bar{d}_j, \quad j = 1, \dots, n, \\ & x_{ij} \geq 0. \end{aligned}$$

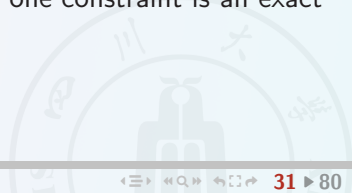
The parameter t changes both feasibility and the optimal shipping plan.

Model Stability Measures Sensitivity to Perturbations

- A **stable** model exhibits small changes in the optimizer and optimal value under small data perturbations.
- An **unstable** or ill-conditioned model can exhibit large changes under small perturbations.
- Near-dependent or redundant constraints may amplify numerical sensitivity.

$$\sum_{i=1}^m \lambda_i g_i(x) = \sum_{i=1}^m \lambda_i b_i \quad \rightarrow \quad \sum_{i=1}^m \lambda_i g_i(x) = \varepsilon + \sum_{i=1}^m \lambda_i b_i.$$

A tiny inconsistent perturbation $\varepsilon \neq 0$ can destroy feasibility when one constraint is an exact linear combination of others.



Matrices Organize Linear Maps and Constraints

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = [a_{ij}] \in \mathbb{R}^{m \times n}.$$

- a_{ij} is the entry in row i and column j .
- $\mathbb{R}^{m \times n}$ is the set of all real $m \times n$ matrices.
- In optimization, rows often represent constraints and columns represent variables.

Transposition Exchanges Rows and Columns

If $A \in \mathbb{R}^{m \times n}$, then $A^T \in \mathbb{R}^{n \times m}$ and

$$A^T = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix}, \quad (A^T)_{ij} = A_{ji}.$$

Useful identities:

$$(A + B)^T = A^T + B^T, \quad (AB)^T = B^T A^T.$$



Column and Row Vectors Have Different Shapes

$$x = \begin{bmatrix} x_1 & \cdots & x_m \end{bmatrix}^T \in \mathbb{R}^{m \times 1},$$
$$y^T = \begin{bmatrix} y_1 & \cdots & y_n \end{bmatrix} \in \mathbb{R}^{1 \times n}.$$

- Scalars are real or complex numbers.
- $\mathbb{R}^m$ denotes Euclidean m -space and is usually identified with $\mathbb{R}^{m \times 1}$.
- Optimization decision variables are conventionally written as column vectors.

Vector Addition and Scalar Multiplication Are Componentwise

For $x, y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$,

$$\begin{aligned}x + y &= [x_1 + y_1, \dots, x_n + y_n]^T, \\ \alpha x &= [\alpha x_1, \dots, \alpha x_n]^T.\end{aligned}$$

These operations satisfy closure, associativity, commutativity of addition, distributivity, and the usual identity and inverse properties.



The Inner Product Induces the Euclidean Norm

$$\langle x, y \rangle = x^T y = \sum_{i=1}^n x_i y_i.$$

- Symmetry: $\langle x, y \rangle = \langle y, x \rangle$.
- Linearity: $\langle \alpha x + \beta y, z \rangle = \alpha \langle x, z \rangle + \beta \langle y, z \rangle$.
- Positive definiteness: $\langle x, x \rangle \geq 0$, with equality iff $x = 0$.
- Euclidean norm:

$$\|x\|_2 = \sqrt{\langle x, x \rangle} = \sqrt{x^T x}.$$



A Subspace Is Closed under Linear Combinations

A nonempty set $S \subseteq \mathbb{R}^n$ is a subspace if

$$x, y \in S, \alpha, \beta \in \mathbb{R} \implies \alpha x + \beta y \in S.$$

For $B \subseteq \mathbb{R}^n$, its linear span is

$$\mathcal{L}(B) = \left\{ \sum_{i=1}^k \alpha_i x_i : \alpha_i \in \mathbb{R}, x_i \in B, k \in \mathbb{N} \right\}.$$

$\mathcal{L}(B)$ collects all finite linear combinations of vectors in B .



Example 5: The Standard Basis Spans $\mathbb{R}^3$

Let

$$B_1 = \{(1, 0, 0)^\top, (0, 1, 0)^\top, (0, 0, 1)^\top\}.$$

Then

$$\begin{aligned}\mathcal{L}(B_1) &= \{x_1(1, 0, 0)^\top + x_2(0, 1, 0)^\top + x_3(0, 0, 1)^\top : x_i \in \mathbb{R}\} \\ &= \{(x_1, x_2, x_3)^\top : x_i \in \mathbb{R}\} = \mathbb{R}^3.\end{aligned}$$

Example 6: Two Vectors Span the x_1 - x_3 Plane

Let

$$B_2 = \{(1, 0, 0)^\top, (1, 0, 1)^\top\}.$$

Then

$$\begin{aligned}\mathcal{L}(B_2) &= \{x_1(1, 0, 0)^\top + x_2(1, 0, 1)^\top : x_1, x_2 \in \mathbb{R}\} \\ &= \{(x_1 + x_2, 0, x_2)^\top : x_1, x_2 \in \mathbb{R}\} \\ &= \{(u, 0, v)^\top : u, v \in \mathbb{R}\}.\end{aligned}$$

Thus $\mathcal{L}(B_2)$ is the x_1 - x_3 plane.

Example 6: Adding a Dependent Vector Does Not Enlarge the Span

Let

$$B_3 = \{(1, 0, 0)^T, (1, 0, 1)^T, (4, 0, 1)^T\}.$$

Because

$$(4, 0, 1)^T = 3(1, 0, 0)^T + (1, 0, 1)^T,$$

the third vector is already in $\mathcal{L}(B_2)$. Therefore

$$\mathcal{L}(B_3) = \mathcal{L}(B_2) = \{(u, 0, v)^T : u, v \in \mathbb{R}\}.$$



The Span Is the Smallest Subspace Containing a Set

- $\mathcal{L}(B)$ is a subspace of $\mathbb{R}^n$.
- If $S = \mathcal{L}(B)$, then B spans S .
- Any intersection of subspaces of $\mathbb{R}^n$ is also a subspace.
- The span can be characterized by

$$\mathcal{L}(B) = \bigcap \{S : S \text{ is a subspace of } \mathbb{R}^n \text{ and } B \subseteq S\}.$$

Hence $\mathcal{L}(B)$ is the unique smallest subspace containing B .

Example 7: Solve the Constraint to Find a Spanning Set

Let

$$S = \{(x_1, x_2, x_3)^T : x_1 + x_2 = 0\}.$$

Since $x_2 = -x_1$,

$$\begin{aligned} S &= \{(x_1, -x_1, x_3)^T : x_1, x_3 \in \mathbb{R}\} \\ &= \{x_1(1, -1, 0)^T + x_3(0, 0, 1)^T : x_1, x_3 \in \mathbb{R}\} \\ &= \mathcal{L}\{(1, -1, 0)^T, (0, 0, 1)^T\}. \end{aligned}$$

Thus S is a two-dimensional subspace.



A Basis Is Independent and Spans the Space

A set B is linearly dependent if there exist distinct vectors $x_1, \dots, x_k \in B$ and scalars, not all zero, such that

$$\alpha_1 x_1 + \dots + \alpha_k x_k = 0.$$

Otherwise B is linearly independent.

Basis

B is a basis for a subspace S if:

- 1 $\mathcal{L}(B) = S$;
- 2 B is linearly independent.



Dimension Counts the Vectors in Any Basis

For a finite-dimensional subspace S :

- every basis of S has the same number of vectors, denoted $\dim(S)$;
- every finite spanning set contains a basis;
- every linearly independent set can be extended to a basis;
- more than $\dim(S)$ vectors in S must be linearly dependent;
- if $S \subseteq T$, then $\dim(S) \leq \dim(T)$.



Rows and Columns Are Vectors in Different Spaces

For $A = [a_{ij}] \in \mathbb{R}^{m \times n}$:

$$A^{(j)} = (a_{1j}, \dots, a_{mj})^T \in \mathbb{R}^m,$$

the j -th column,

$$A_{(i)} = (a_{i1}, \dots, a_{in}) \in \mathbb{R}^{1 \times n},$$

the i -th row.

The same matrix can therefore be viewed as

$$A = [A^{(1)} \ \dots \ A^{(n)}] \quad \text{or as a stack of row vectors.}$$



Row Rank Equals Column Rank

- Column space: $\text{col}(A) = \mathcal{L}\{A^{(1)}, \dots, A^{(n)}\} \subseteq \mathbb{R}^m$.
- Row space: $\text{row}(A) = \mathcal{L}\{A_{(1)}, \dots, A_{(m)}\} \subseteq \mathbb{R}^n$.
- The fundamental rank theorem gives

$$\text{rank}(A) = \dim \text{col}(A) = \dim \text{row}(A).$$

Rank measures the number of independent directions represented by the matrix.



Matrix Operations Extend Scalar Arithmetic

For conformable matrices:

$$A + B = [a_{ij} + b_{ij}],$$

$$\alpha A = [\alpha a_{ij}],$$

$$(AB)_{ij} = \sum_{\ell=1}^k a_{i\ell} b_{\ell j},$$

$$A \in \mathbb{R}^{m \times k}, \quad B \in \mathbb{R}^{k \times n}.$$

Therefore $AB \in \mathbb{R}^{m \times n}$. In general, matrix multiplication is associative and distributive but not commutative.



The Identity Matrix Uses the Kronecker Delta

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = [\delta_{ij}] \in \mathbb{R}^{n \times n},$$

where the Kronecker delta is

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

For every conformable matrix A , $AI = IA = A$.



A Submatrix Selects Rows and Columns from a Matrix

Consider

$$A = \begin{bmatrix} 6 & 1 & 3 & 4 & 0 & 5 \\ 2 & 1 & 4 & 0 & 0 & 1 \\ 1 & 0 & 1 & -1 & 1 & 0 \end{bmatrix}.$$

Selecting rows $\{2, 3\}$ and columns $\{3, 4, 6\}$ gives

$$B = \begin{bmatrix} 4 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}.$$

Submatrices support block algebra, factorization, and decomposition arguments.

Partitioned Matrices Treat Submatrices as Blocks

The matrix from the previous slide can be partitioned as

$$A = \left[\begin{array}{cc|cc|c|c} 6 & 1 & 3 & 4 & 0 & 5 \\ 2 & 1 & 4 & 0 & 0 & 1 \\ \hline 1 & 0 & 1 & -1 & 1 & 0 \end{array} \right] = \begin{bmatrix} B & D & O & b \\ -c_B^T & -c_D^T & 1 & 0 \end{bmatrix},$$

with

$$B = \begin{bmatrix} 6 & 1 \\ 2 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 3 & 4 \\ 4 & 0 \end{bmatrix}, \quad O = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad b = \begin{bmatrix} 5 \\ 1 \end{bmatrix}.$$

Block Multiplication Follows the Same Conformability Rules

Let

$$A = \begin{bmatrix} C & D \\ E & F \end{bmatrix}, \quad B = \begin{bmatrix} G & H & J \\ K & L & M \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} CG + DK & CH + DL & CJ + DM \\ EG + FK & EH + FL & EJ + FM \end{bmatrix}.$$

Each block product must be dimensionally conformable.



A Nonsingular Matrix Has a Two-Sided Inverse

For $A, B \in \mathbb{R}^{n \times n}$,

$$AB = BA = I_n$$

means that A is nonsingular and $B = A^{-1}$.

- The inverse, when it exists, is unique.
- A matrix without an inverse is singular.
- $(AB)^{-1} = B^{-1}A^{-1}$ for nonsingular A and B .



Full Rank Characterizes Nonsingularity

For a square matrix $A \in \mathbb{R}^{n \times n}$, the following are equivalent:

A is nonsingular $\iff \text{rank}(A) = n \iff$ columns of A are independent.

Compare

$$\underbrace{\begin{bmatrix} 1 & 1 & 4 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}}_{\text{singular}} \quad \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}}_{\text{nonsingular}}.$$

Elementary Row Operations Preserve the Row Space

The three elementary row operations are:

- 1 scale a row by $\alpha \neq 0$;
- 2 replace row i by row $i + \alpha$ row j ;
- 3 exchange two rows.

To compute an inverse, row reduce the augmented matrix:

$$[A \mid I_n] \longrightarrow [I_n \mid A^{-1}].$$

Failure to obtain I_n on the left means that A is singular.



A Block-Triangular Matrix Has a Simple Inverse

Let $B \in \mathbb{R}^{m \times m}$ be nonsingular. Then

$$\begin{bmatrix} B & 0 \\ -c_B^\top & 1 \end{bmatrix}^{-1} = \begin{bmatrix} B^{-1} & 0 \\ c_B^\top B^{-1} & 1 \end{bmatrix}.$$

Verification follows by block multiplication:

$$\begin{bmatrix} B^{-1} & 0 \\ c_B^\top B^{-1} & 1 \end{bmatrix} \begin{bmatrix} B & 0 \\ -c_B^\top & 1 \end{bmatrix} = I_{m+1}.$$



The Same Row Operations Extend to a Block System

Suppose row operations transform

$$[B \mid I_m] \longrightarrow [I_m \mid B^{-1}].$$

Applying them to the upper block row gives

$$\left[\begin{array}{cc|cc} B & 0 & I_m & 0 \\ -c_B^T & 1 & 0^T & 1 \end{array} \right] \longrightarrow \left[\begin{array}{cc|cc} I_m & 0 & B^{-1} & 0 \\ -c_B^T & 1 & 0^T & 1 \end{array} \right].$$

Adding c_B^T times the block row to the last row yields the inverse on the right.



Example 8: Solve a Linear System

Solve $Ax = b$ for

$$A = \begin{bmatrix} 1 & 1 & -1 \\ -2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}.$$

We can row reduce $[A \mid I \mid b]$ to obtain both A^{-1} and $x = A^{-1}b$.



Example 8: Gauss-Jordan Elimination Gives A^{-1} and x

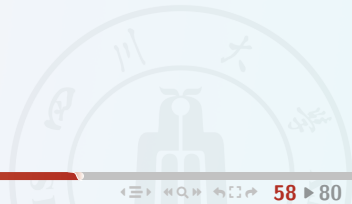
Row reduction produces

$$[A \mid I_3 \mid b] \longrightarrow [I_3 \mid A^{-1} \mid x],$$

where

$$A^{-1} = \begin{bmatrix} 0 & -\frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}, \quad x = \begin{bmatrix} -\frac{2}{3} \\ \frac{7}{6} \\ -\frac{1}{2} \end{bmatrix}.$$

Direct substitution confirms $Ax = b$.



Elementary Matrices Encode Row Operations

- The augmented matrix for $Ax = b$ is $[A \mid b]$.
- Gauss-Jordan elimination transforms it to $[I \mid A^{-1}b]$ when A is nonsingular.
- Performing a row operation on I produces an elementary matrix E .

For example,

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$$

Left multiplication EA applies the corresponding row operation to A .

Gaussian Elimination Separates Forward and Backward Steps

- **Forward elimination:** transform $[A \mid b]$ to $[U \mid \bar{b}]$, where U is upper triangular.
- **Back substitution:** solve the triangular system from the last variable upward.

Example system:

$$\begin{aligned}2x_2 - x_3 &= 8, \\x_1 + 2x_2 &= 6, \\3x_2 + 4x_3 &= 12.\end{aligned}$$

Gaussian Elimination Produces an Upper-Triangular System

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & 8 \\ 1 & 2 & 0 & 6 \\ 0 & 3 & 4 & 12 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 6 \\ 0 & 2 & -1 & 8 \\ 0 & 3 & 4 & 12 \end{array} \right]$$

$$\xrightarrow{R_2 \leftrightarrow R_3, R_3 \leftarrow R_3 - \frac{2}{3}R_2} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 6 \\ 0 & 3 & 4 & 12 \\ 0 & 0 & -\frac{11}{3} & 0 \end{array} \right].$$

Back substitution gives $x_3 = 0$, $x_2 = 4$, and $x_1 = -2$.



Elimination Is Multiplication by Elementary Matrices

For the preceding system,

$$U = E_3 E_2 E_1 A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & -\frac{11}{3} \end{bmatrix}.$$

Here:

- E_1 exchanges the first two rows;
- E_2 exchanges the last two rows;
- E_3 eliminates the entry below the second pivot.

Thus a row-reduction sequence is an explicit matrix factorization.



Pivoting Controls the Row-Echelon Process

- A row-echelon form has each pivot to the right of the pivot above it.
- For a nonsingular square matrix, elimination produces an upper-triangular matrix.
- A **pivot** is the leading nonzero entry used to eliminate entries below it.
- **Pivoting** exchanges rows to select a suitable nonzero pivot.
- A **permutation matrix** is obtained by permuting the rows of I and records these exchanges.



LU Factorization Reuses Gaussian Elimination

With row pivoting, Gaussian elimination yields

$$PA = LU,$$

where P is a permutation matrix, L is lower triangular, and U is upper triangular. To solve $Ax = b$:

- 1 solve $Ly = Pb$ by forward substitution;
- 2 solve $Ux = y$ by back substitution.

The factorization is especially useful when many right-hand sides share the same matrix A .

Coefficient Restrictions Define Four Kinds of Combinations

For $x_1, \dots, x_k \in \mathbb{R}^n$:

Combination	Expression	Coefficient restriction
Linear	$\sum_i \alpha_i x_i$	$\alpha_i \in \mathbb{R}$
Affine	$\sum_i \alpha_i x_i$	$\sum_i \alpha_i = 1$
Conical	$\sum_i \alpha_i x_i$	$\alpha_i \geq 0$
Convex	$\sum_i \alpha_i x_i$	$\alpha_i \geq 0, \sum_i \alpha_i = 1$

Each added restriction changes the geometry of the generated set.



Linear and Affine Hulls Generate the Smallest Matching Sets

For nonempty $S \subseteq \mathbb{R}^n$:

$$\mathcal{L}(S) = \left\{ \sum_{i=1}^k \alpha_i x_i : \alpha_i \in \mathbb{R}, x_i \in S, k \in \mathbb{N} \right\},$$

$$\mathcal{A}(S) = \left\{ \sum_{i=1}^k \alpha_i x_i : \alpha_i \in \mathbb{R}, \sum_{i=1}^k \alpha_i = 1, x_i \in S \right\}.$$

$\mathcal{L}(S)$ is the smallest subspace containing S ; $\mathcal{A}(S)$ is the smallest affine set containing S .

Conical and Convex Hulls Add Nonnegativity

$$\mathcal{C}(S) = \left\{ \sum_{i=1}^k \alpha_i x_i : \alpha_i \geq 0, x_i \in S \right\},$$
$$\mathcal{X}(S) = \left\{ \sum_{i=1}^k \alpha_i x_i : \alpha_i \geq 0, \sum_{i=1}^k \alpha_i = 1, x_i \in S \right\}.$$

- $\mathcal{C}(S)$ is the smallest convex cone containing S .
- $\mathcal{X}(S) = \text{conv}(S)$ is the smallest convex set containing S .

Example 9: Two Basis Vectors Generate Four Different Hulls

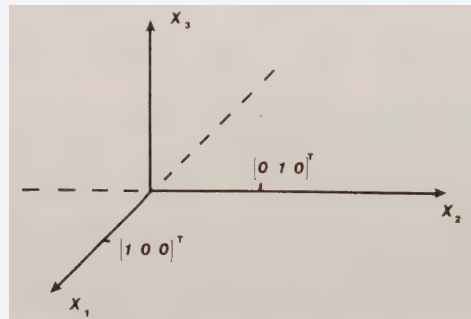
Let $S = \{e_1, e_2\} \subset \mathbb{R}^3$.

$$\mathcal{L}(S) = \{(u, v, 0)^T : u, v \in \mathbb{R}\},$$

$$\mathcal{A}(S) = \{(u, v, 0)^T : u + v = 1\},$$

$$\mathcal{C}(S) = \{(u, v, 0)^T : u, v \geq 0\},$$

$$\mathcal{X}(S) = \{(u, v, 0)^T : u, v \geq 0, u + v = 1\}.$$



Hulls Characterize the Corresponding Closed Operations

For nonempty $S \subseteq \mathbb{R}^n$:

- $S \subseteq \mathcal{X}(S) \subseteq \mathcal{A}(S) \subseteq \mathcal{L}(S)$;
- $S \subseteq \mathcal{X}(S) \subseteq \mathcal{C}(S) \subseteq \mathcal{L}(S)$;
- S is a subspace iff $S = \mathcal{L}(S)$;
- S is affine iff $S = \mathcal{A}(S)$;
- S is a convex cone iff $S = \mathcal{C}(S)$;
- S is convex iff $S = \mathcal{X}(S)$.

Geometrically: affine sets contain full lines, convex cones contain nonnegative combinations, and convex sets contain line segments.



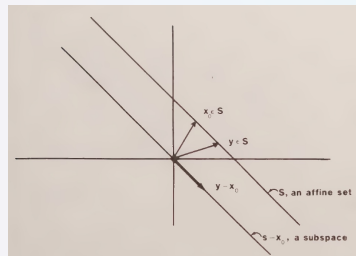
An Affine Set Is a Translated Subspace

A nonempty set $S \subseteq \mathbb{R}^n$ is affine iff, for any $x_0 \in S$,

$$S - x_0 = \{x - x_0 : x \in S\}$$

is a linear subspace.

- $S = x_0 + V$ for a subspace V ;
- $\dim(S) = \dim(V)$;
- for a set T , $\dim(T)$ is defined as $\dim \mathcal{A}(T)$.



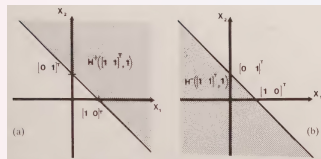
A Hyperplane Separates Two Halfspaces

For $c \neq 0$ and $\beta \in \mathbb{R}$:

$$\mathcal{H}(c, \beta) = \{x : c^T x = \beta\},$$

$$\mathcal{H}^+(c, \beta) = \{x : c^T x \geq \beta\},$$

$$\mathcal{H}^-(c, \beta) = \{x : c^T x \leq \beta\}.$$

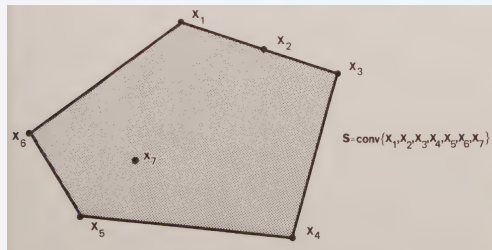


A Simplex Is the Basic Affinely Independent Polytope

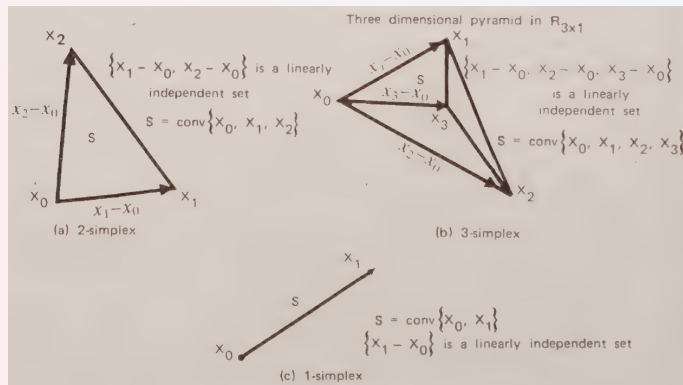
- A convex polytope is the convex hull of finitely many points:

$$P = \mathcal{X}\{x_1, \dots, x_k\}.$$

- $\{x_0, \dots, x_k\}$ is affinely independent iff $\{x_1 - x_0, \dots, x_k - x_0\}$ is linearly independent.
- A k -simplex is the convex hull of $k + 1$ affinely independent points.



Segments, Triangles, and Tetrahedra Are Simplices



A 1-simplex is a segment, a 2-simplex is a triangle, and a 3-simplex is a tetrahedron.

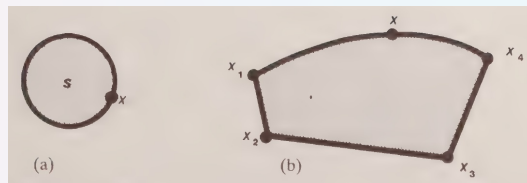
Extreme Points Cannot Be Written as Nontrivial Convex Mixtures

A point $z \in S$, with S convex, is extreme if

$$z = \alpha x + (1 - \alpha)y, \quad x, y \in S, \quad 0 < \alpha < 1$$

implies $x = y = z$.

- Every vertex of a simplex is an extreme point.
- An extreme point is not in the relative interior of a segment joining two distinct points of S .



A Linear Program Can Be Written Componentwise

$$\begin{aligned} \min_x \quad & c_1x_1 + \cdots + c_nx_n \\ \text{subject to} \quad & a_{i1}x_1 + \cdots + a_{in}x_n \leq b_i, \quad i = 1, \dots, m, \\ & x_j \geq 0, \quad j = 1, \dots, n. \end{aligned}$$

Componentwise notation exposes the role of each variable and each constraint.



Matrix Notation Reveals the Feasible Geometry

The same problem is

$$\min_x c^T x \quad \text{subject to} \quad Ax \leq b, \quad x \geq 0.$$

Since

$$Ax = A^{(1)}x_1 + \cdots + A^{(n)}x_n,$$

the feasible set is

$$F = \{x \in \mathbb{R}^n : Ax \leq b, \quad x \geq 0\}.$$



The Feasible Set of a Linear Program Is Convex

Let $x_1, x_2 \in F$ and $\alpha \in [0, 1]$. For

$$z = \alpha x_1 + (1 - \alpha)x_2,$$

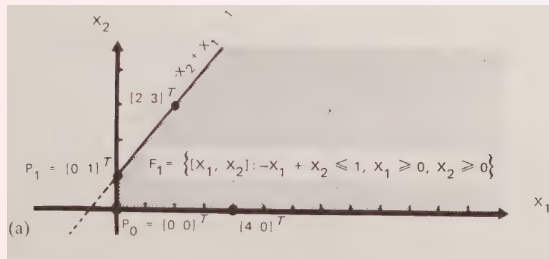
linearity gives

$$Az = \alpha Ax_1 + (1 - \alpha)Ax_2 \leq b, \quad z \geq 0.$$

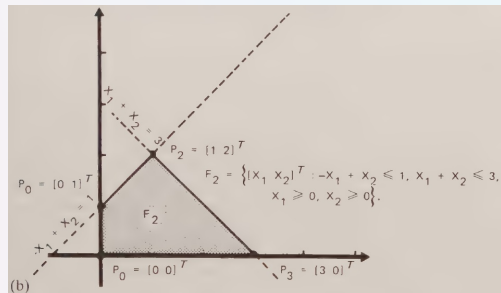
Hence $z \in F$.

- A polyhedron is an intersection of finitely many closed halfspaces.
- Every polytope is a bounded polyhedron.
- Every bounded polyhedron is a polytope; an unbounded polyhedron is not.

Boundedness Distinguishes Polytopes from General Polyhedra



Unbounded polyhedron: not a polytope.



Bounded polyhedron: a polytope.

Minkowski-Weyl Gives a Finite Description of Every Polyhedron

Finite generation theorem

A nonempty polyhedron $F \subseteq \mathbb{R}^n$ can be represented using finite sets of points $\{x_1, \dots, x_d\}$ and rays $\{y_1, \dots, y_\ell\}$:

$$F = \mathcal{X}\{x_1, \dots, x_d\} + \mathcal{C}\{y_1, \dots, y_\ell\}.$$

Equivalently,

$$x = \sum_{i=1}^d \alpha_i x_i + \sum_{j=1}^{\ell} \beta_j y_j, \quad \alpha_i, \beta_j \geq 0, \quad \sum_{i=1}^d \alpha_i = 1.$$



A Polyhedron Combines a Bounded Base with Recession Rays

For the illustrated set,

$$F_2 = \mathcal{X}\{p_0, p_1\} + \mathcal{C}\{v_0, v_1\}.$$

Thus every $x \in F_2$ has the form

$$x = \alpha_0 p_0 + \alpha_1 p_1 + \beta_0 v_0 + \beta_1 v_1,$$

where $\alpha_0 + \alpha_1 = 1$ and all coefficients are nonnegative.

